

Lecture 14 - Resolving Conflicting Design Requirements - Developing a strategy

Prof James Busfield



- Conflicting objectives
- Multi-objective optimisation
- Reaching a compromise
- Value functions and exchange constants
- Case studies:

Air tanks for truck

Casing for mobile phone

Resources:

- “Multi-objective optimisation in material design and selection”, by M.F. Ashby, Acta Mater. (2000), vol 48, p 359 - 369.
- The Cambridge Material Selector software -- Granta Design, Cambridge (www.grantadesign.com)

- Real life often requires a compromise between conflicting objectives:
 - Price versus performance of a bike or car
 - Accumulating wealth versus quality of life
 - Health versus the pleasures of rich food.
- Conflict arises because the choice that optimises one metric of performance will not in general do the same for the others.
- Best choice is a compromise, optimising none but pushing all as close to optimum as their interdependence allows.

- Common design objectives, influencing choice of material, are:

Minimising **mass** (sprint bike; satellite components)

Minimising **volume** (mobile phone; minidisk player)

Maximising **energy density** (flywheels, springs)

Minimising **eco-impact** (carbon footprint)

Minimising **cost** (everything)

Objectives

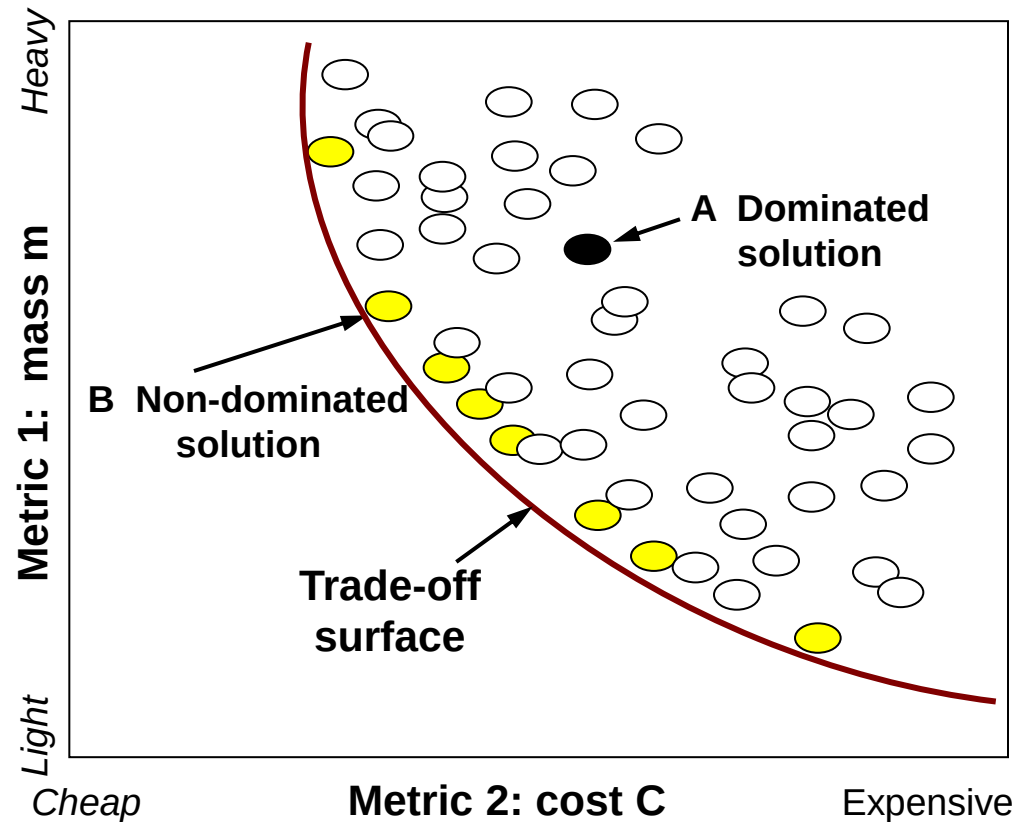
- Each objective defines a performance metric. Take, as example

$\left. \begin{array}{l} \text{mass, } m \\ \text{cost, } C \end{array} \right\}$ we wish to minimise both,
(all other constraints being met)

- Solutions that minimise mass seldom minimise cost, and *vice versa*

Multi-objective Optimisation: Terminology

- **Solution:** a viable choice, meeting constraints, but not necessarily optimum by either criterion.
- **Dominated solution:** one that is unambiguously non-optimal (as **A**: some other solution is better by both metrics)
- **Non-dominated solution:** one that is optimal by one metric (as **B**: optimal by one criterion but not necessarily by both)
- **Trade-off surface:** the surface on which the non-dominated solutions lie (also called the Pareto Front)
- **There are three strategies** for finding best compromise (on the next 3 pages)

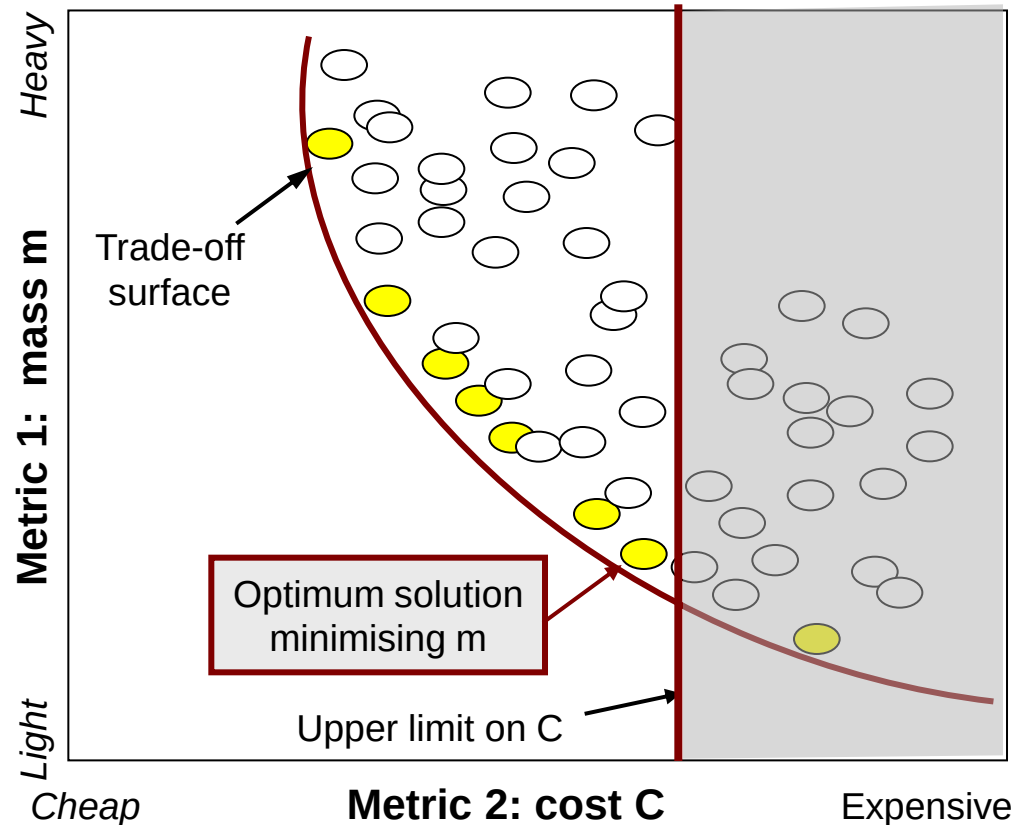


Finding a Compromise: Strategy 1

- **Make trade-off plot**
- **Sketch trade-off surface**
- **Reformulate** all but one of the objectives as constraints, setting an upper limit for it

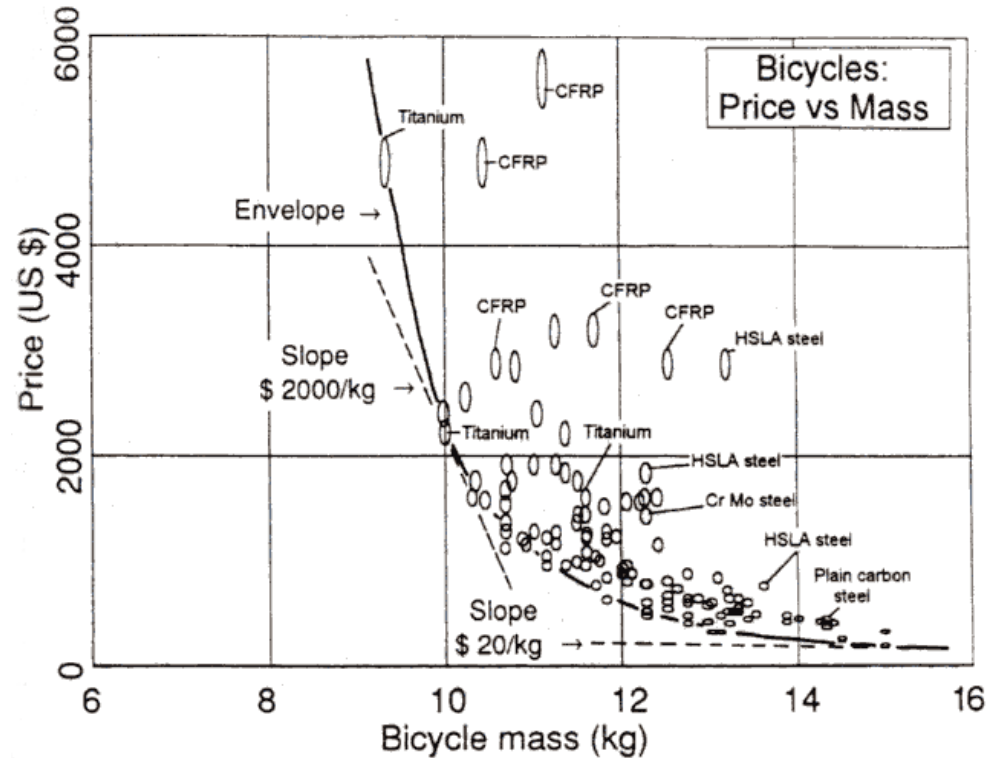
Mass and cost of bicycles:

- Good if you have budget limit
- Trade-off surface leads you to the best choice within budget
- But this is not a true optimisation
 - cost has been treated as a constraint, not an objective.



Finding a Compromise: Strategy 2

- **Make trade-off plot**
- **Sketch trade-off surface**
- **Use intuition** to select a solution on the trade-off surface



Mass and cost of bicycles:

- A very well defined trade off surface.
- *Solutions* on or near the surface offer the best compromise between mass and cost.
- Choose from among these; the choice depends on how highly you value a light bicycle. A question of their *relative value*

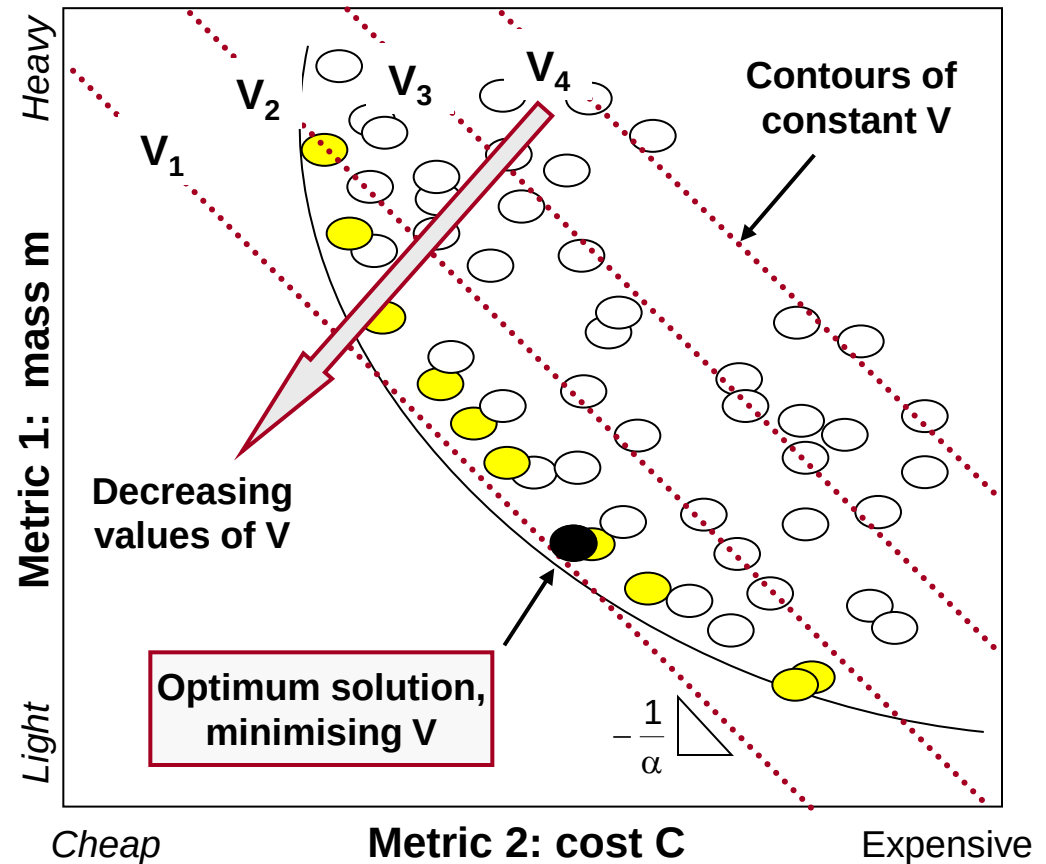
Finding a Compromise: Strategy 3

Define locally linear
value function V

$$V = \alpha m + C$$

Seek the option with
smallest V :

- Either **evaluate** V for each solution, and rank,
- or **make tradeoff plot**
- **plot on it contours of** V
- lines of constant V have slope $-1/\alpha$
- **Read off** the solution with lowest V



Trade-offs: choosing a fridge



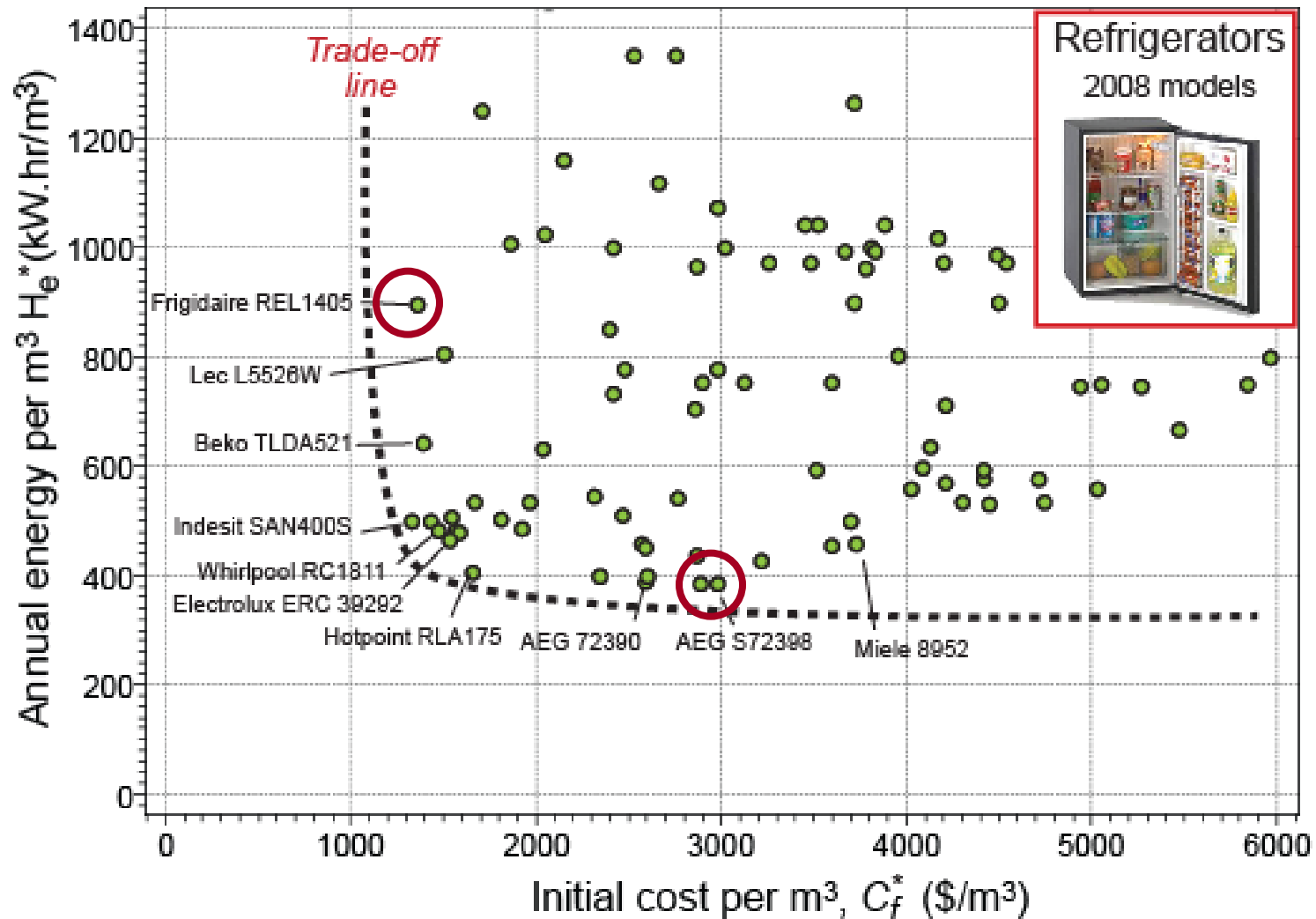
Classic trade off: initial cost vs. running cost

Metric 1: Initial cost of fridge (per m³) C_f^* \$/m³

Metric 2: Annual energy (per m³) H_e^* kWhr/m³

Trade-offs: choosing a fridge 1

Classic trade off: initial cost vs. running cost



Trade-offs: choosing a fridge 2

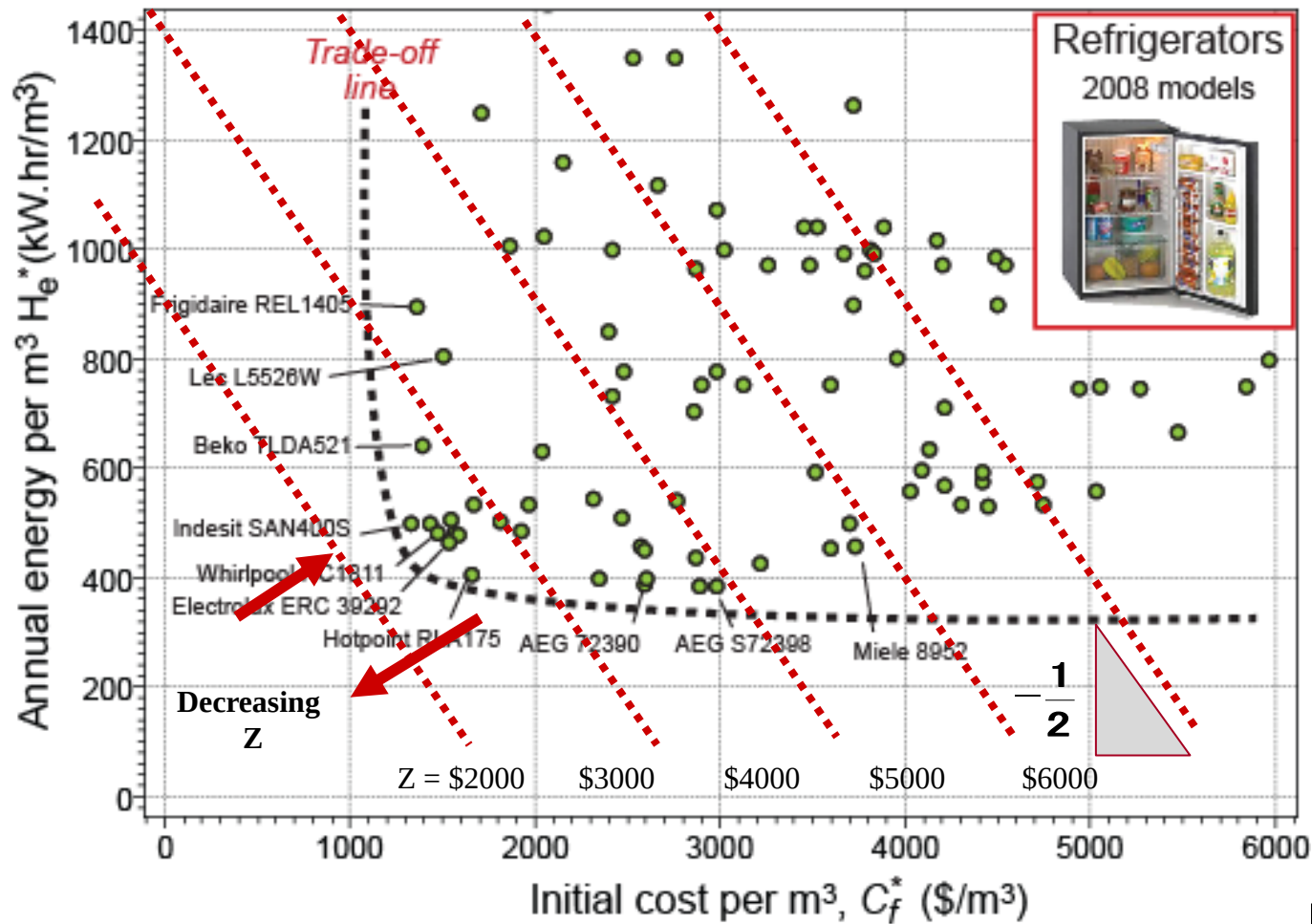
Define Z as
life cost / m³

$$Z = \alpha_e H_e t + C_f$$

$$Z = 2H_e + C_f$$

α_e = energy cost (\$0.2 / kW.hr)

t = life for fridge (10 years)



Materials for transport systems



Steel



Steel / Alu



Aluminum



Al/Ti/Composites



Composites

Choice of material depends on system

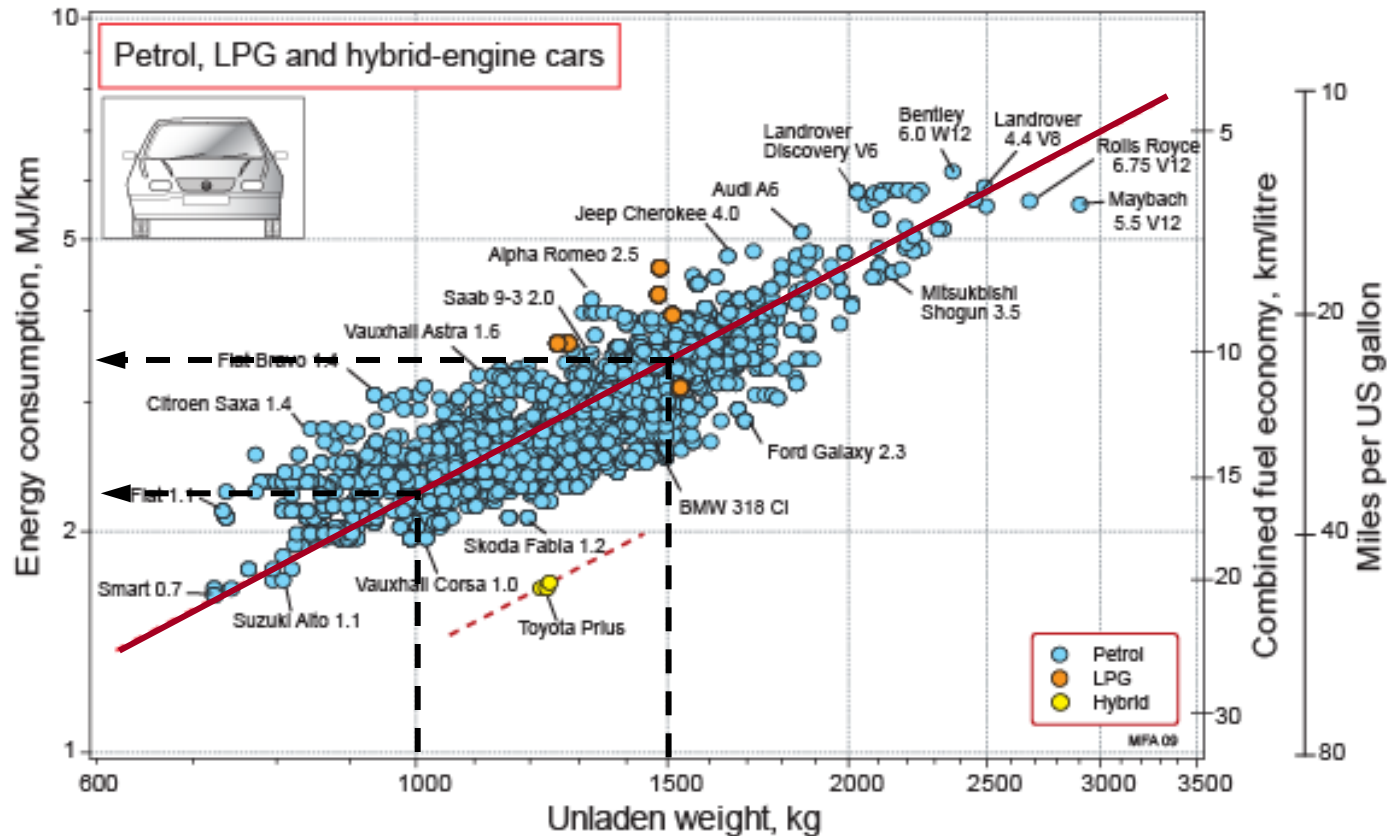
- **Mass**, in transport systems, means **fuel**
- **Life cost** = initial cost, **C** + fuel cost over life, scaling with mass **m**
- **Penalty function** $Z = C + \alpha m$

For a beam: $C \propto \frac{C_m \rho}{\sigma_y^{2/3}}$ $m \propto \frac{\rho}{\sigma_y^{2/3}}$

Penalty function: $Z = C + \alpha m = \frac{\rho}{\sigma_y^{2/3}} \left(C_m + \alpha \right)$

- Must first establish **exchange constant**, α

Transport: mass vs. cost



- Energy (in MJ/km) = 2.3×10^{-3} m (in kg)
- Gasoline energy = 35 MJ/liter
- Gasoline cost = \$ 0.5 / liter (USA)
- Life km = 200,000
- Exchange constant α (the life-value of saving 1 kg) = \$5/kg

The exchange constant α for transport

Exchange constants for transport systems

Transport system	α (\$ per kg)
Family car	1.5 to 5
Truck	5 to 20
Civil aircraft	100 to 500
Military hardware	500 to 2000
Space vehicle	3000 to 10,000



$$V = \alpha m + C$$

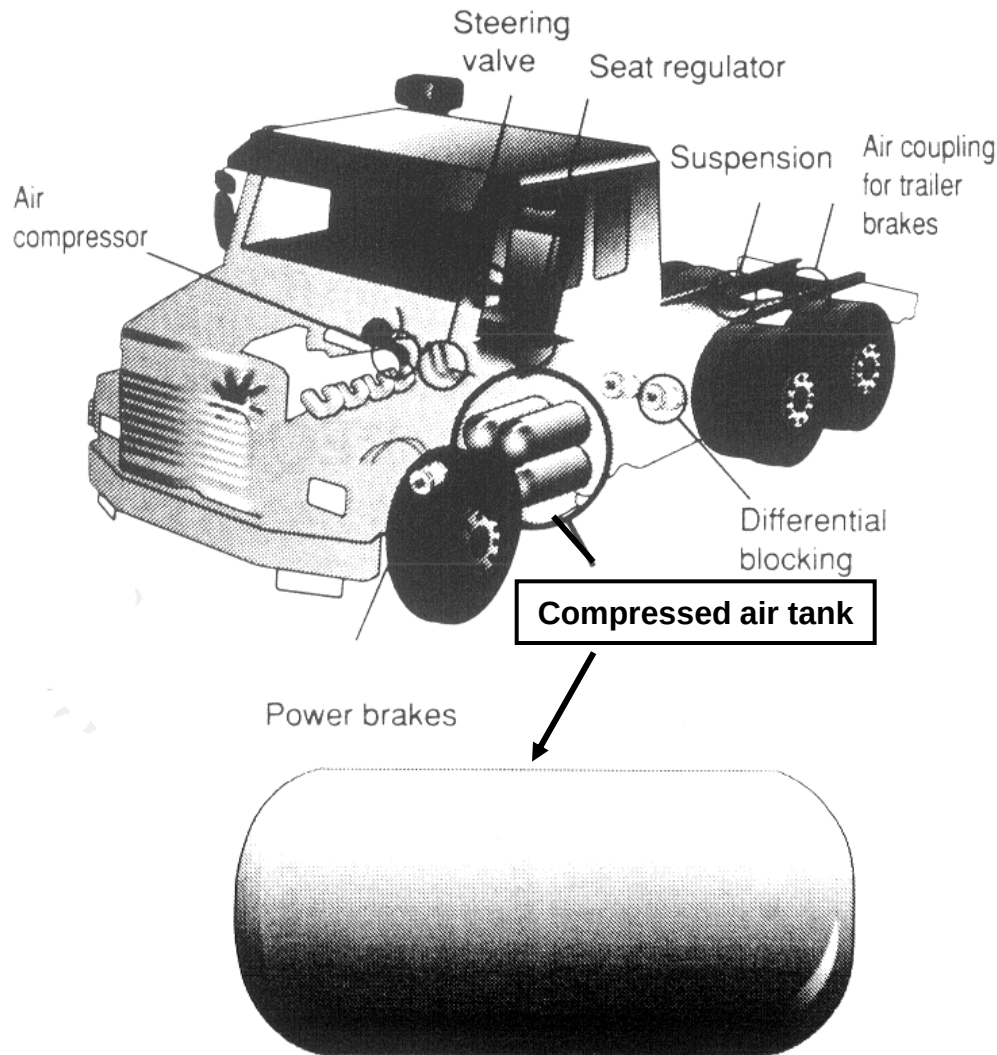


$$\alpha = \left(\frac{\partial V}{\partial m} \right)_C$$

The quantity α is called an “exchange constant” -- it measures the **value of performance**, here the value of saving 1 kg of mass.

Case study: Air Cylinders for Trucks

Design goal: lighter, cheap air cylinders for trucks.



Design Requirements for the Air Cylinder

Specification

Function

Pressure vessel

Objectives

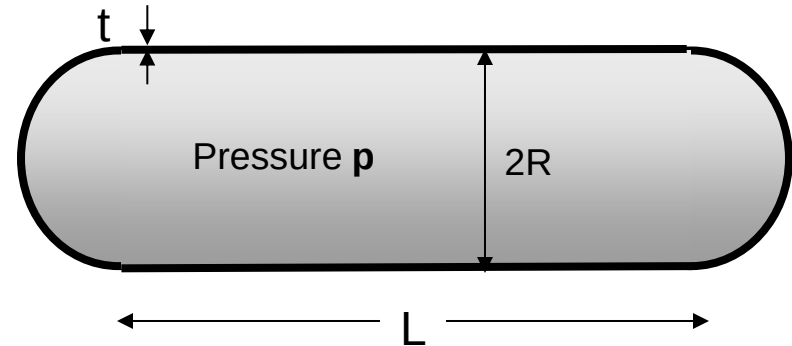
- Minimise mass
- Minimise cost

Constraints

- Dimensions L , R , pressure p , given
- Must not corrode in water or oil
- Working temperature -50 to $+100^{\circ}\text{C}$
- Safety: must not fail by yielding
- Adequate toughness: $K_{1c} > 15 \text{ MPa.m}^{1/2}$

Free variables

- Wall thickness, t ;
- Choice of material



R = radius
 L = length
 ρ = density
 p = pressure
 t = wall thickness

Analysis of the Air Cylinder

Vol of material in cylinder wall

Objective 1

$$m = (2\pi R L t + 4\pi R^2 t) \rho$$
$$= 2\pi R L t \left(1 + \frac{2R}{L} \right)$$

Aspect ratio Q

Constraint

$$\sigma = \frac{pR}{t} < \frac{\sigma_y}{S}$$

Eliminate t to give:

Metric 1

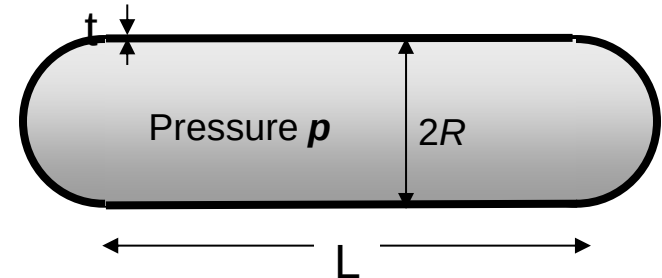
$$m = 2\pi R^2 L (1+Q) p S \left[\frac{\rho}{\sigma_y} \right]$$

Objective 2

$$C = C_m m$$

Metric 2

$$C = 2\pi R^2 L (1+Q) p S \left[\frac{C_m \rho}{\sigma_y} \right]$$



R = radius

L = length

ρ = density

p = pressure

t = wall thickness

σ_y = yield strength

S = safety factor

Q = aspect ratio $2R/L$

This is a problem of substitution. The tank is currently made of a plain carbon steel.

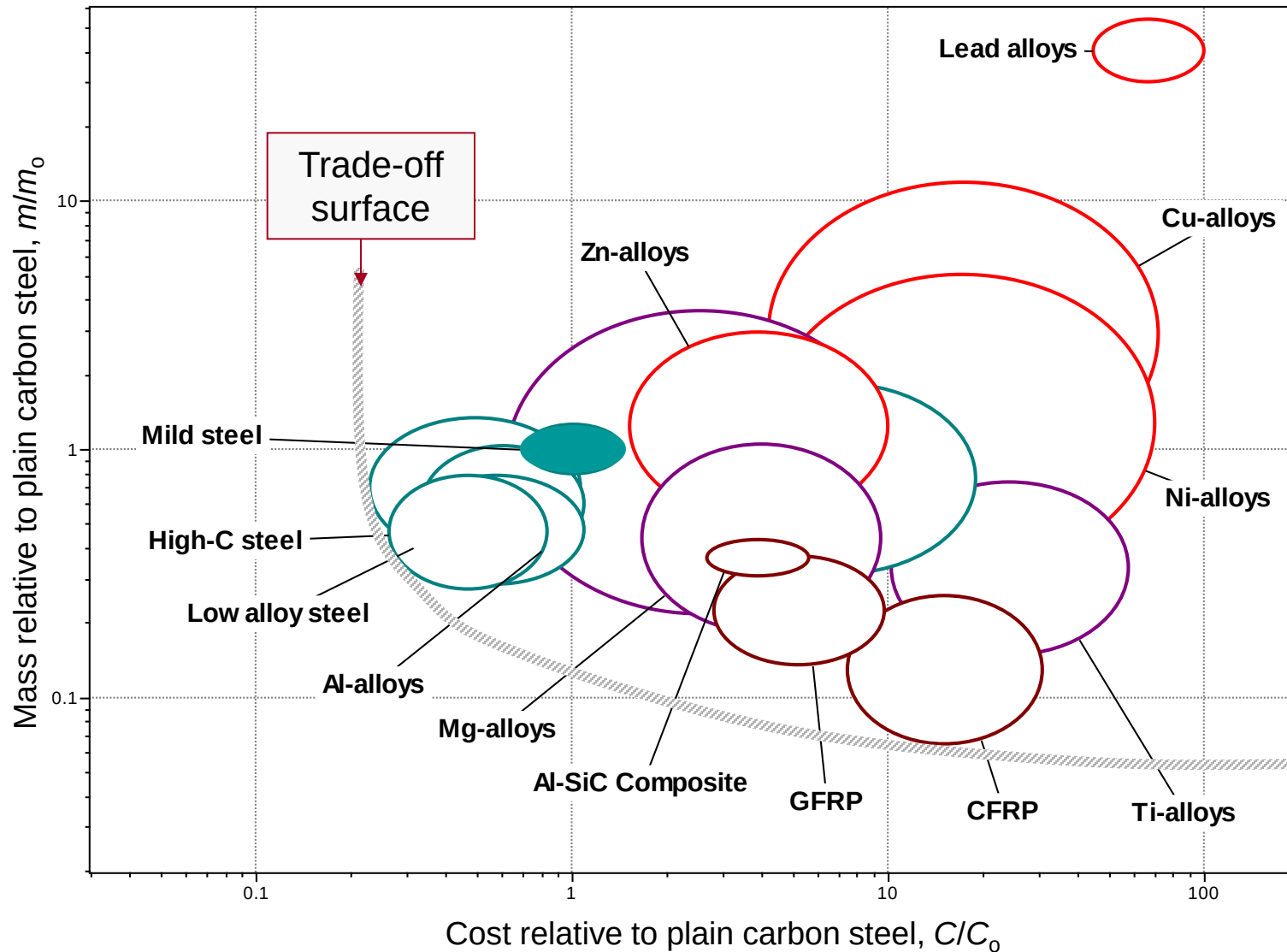
The mass m and cost C of a tank made from an alternative material M , differs (for the same strength) from one made of M_o by the factors

$$\frac{m}{m_o} = \left(\frac{\rho}{\sigma_y} \right) \cdot \left(\frac{\sigma_{y,o}}{\rho_o} \right) \quad \frac{C}{C_o} = \left(\frac{C_m \rho}{\sigma_y} \right) \cdot \left(\frac{\sigma_{y,o}}{C_{m,o} \rho_o} \right)$$

For plain carbon steel $\rho_o / \sigma_{y,o} = 0.03$ and $C_{m,o} \rho_o / \sigma_{y,o} = 0.01$

Explore the trade-off between $\frac{m}{m_o}$ and $\frac{C}{C_o}$

Trade-off plot



Additional constraints:

$$K_{1c} > 15 \text{ MPa.m}^{1/2}$$

$$T_{\max} > 373 \text{ K}$$

$$T_{\min} < 223 \text{ K}$$

Water: good +

Organics: good +

Finding a Compromise: The Value Function

- Aluminium alloy and low alloy steels offer modest reductions in mass at little or no increase in material cost.
- The lightest solutions are GFRP, CRRP and Titanium alloys, but at a cost penalty -- is it worth it? Define a **relative value function**:

$$V^* = \frac{V}{C_o} = \alpha^* \frac{m}{m_o} + \frac{C}{C_o}$$

- The relative exchange constant, α^* , is related to α from before by

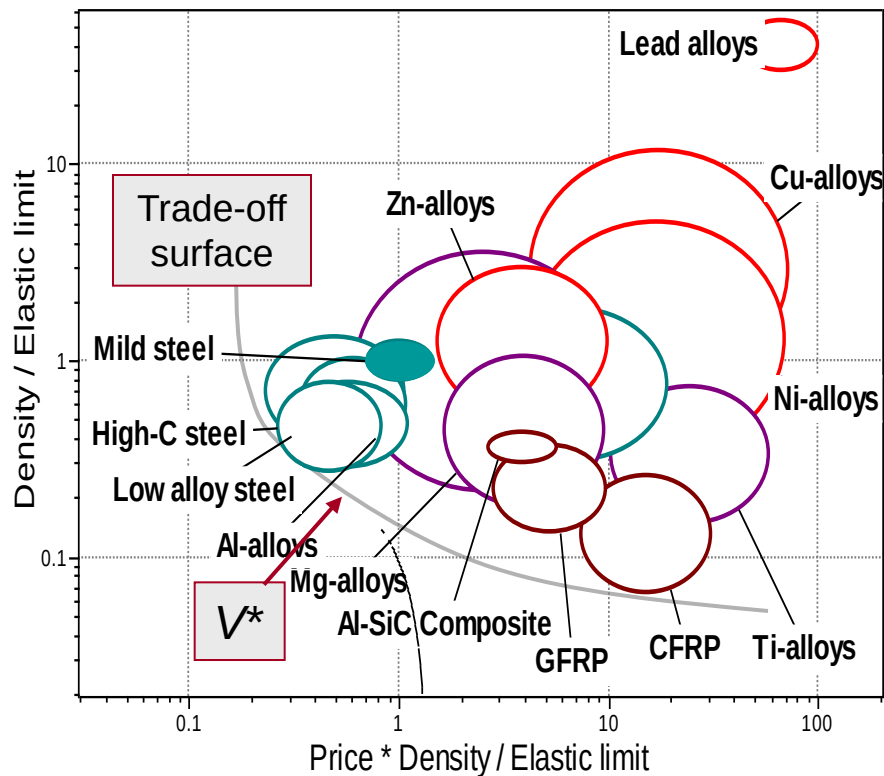
$$\alpha^* = \frac{m_o}{C_o} \alpha \quad (\text{a pure number})$$

- With $m_o = 10$ kg, $C_o = £50$ and $\alpha = £20/\text{kg}$ (trucks), $\alpha^* = 4$.
 - (a) evaluate V^* numerically and rank candidates, or
 - (b) plot onto relative trade-off plot (line of slope $-\frac{1}{4}$)

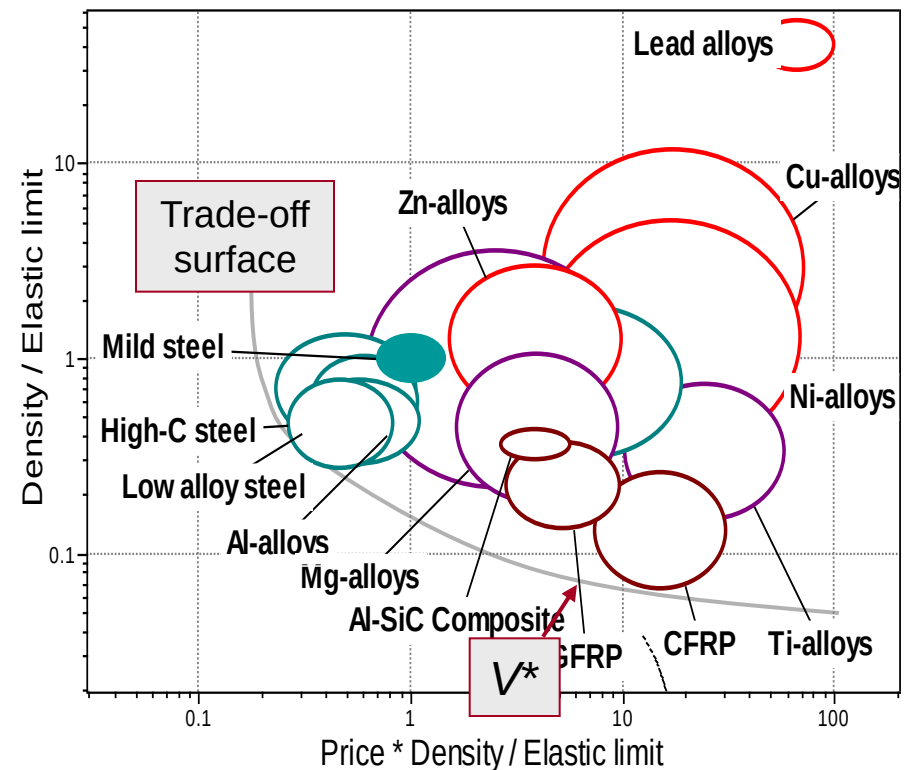
Value Function on a Trade-off Plot

$$V^* = \alpha^* \frac{m}{m_0} + \frac{C}{C_0}$$

Value contour for $\alpha^* = 4$ ($\alpha = \text{£}20/\text{kg}$)



Value contour for $\alpha^* = 200$ ($\alpha = \text{£}1000/\text{kg}$)



- Value lines are curved because of logarithmic scales.

Case Study: Casing for a Mobile Phone or MP3

- Electronic equipment -- portable computers, players, mobile phones -- all miniaturised; many now less than 12 mm thick
- An ABS or Polycarbonate casing has to be $> 1\text{mm}$ thick to be stiff enough for protection; casing occupies 20% of the volume



- Find best material for a stiff casing of minimum thickness and weight

Objective 1 minimise casing thickness

Objective 2 minimise casing mass

- The thinnest may not be the lightest. A need to explore the trade-offs

Performance Metrics for the Casing

Function

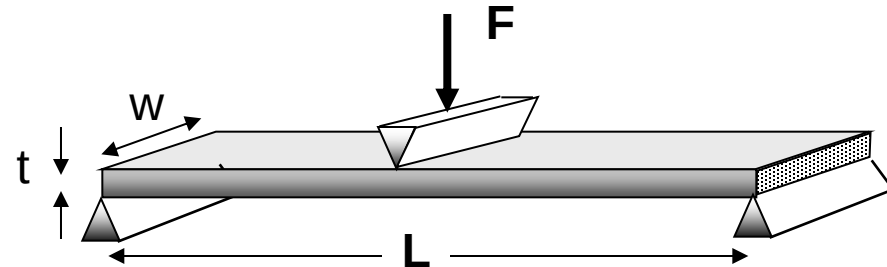
Stiff casing

Constraints

- Stiffness, S

$$S = \frac{48EI}{L^3} \text{ with } I = \frac{wt^3}{12}$$

- Adequate toughness,
 $G_{1c} > 1\text{kJ/m}^2$



Objective 1

Minimise thickness t

Metric 1

$$t = \left(\frac{SL^3}{4Ew} \right)^{1/3} \propto \frac{1}{E^{1/3}}$$

Objective 2

Minimise mass m

Metric 2

$$m = \left(\frac{12 S w^2}{C} \right)^{1/3} L^2 \left(\frac{\rho}{E^{1/3}} \right) \propto \frac{\rho}{E^{1/3}}$$

m = mass

w = width

L = length

ρ = density

t = thickness

S = required stiffness

I = second moment of area

E = Young's Modulus

This is a substitution problem. Suppose the casing is currently made of a material M_o .

The thickness of a casing made from an alternative material M , differs (for the same stiffness) from one made of M_o by the factor

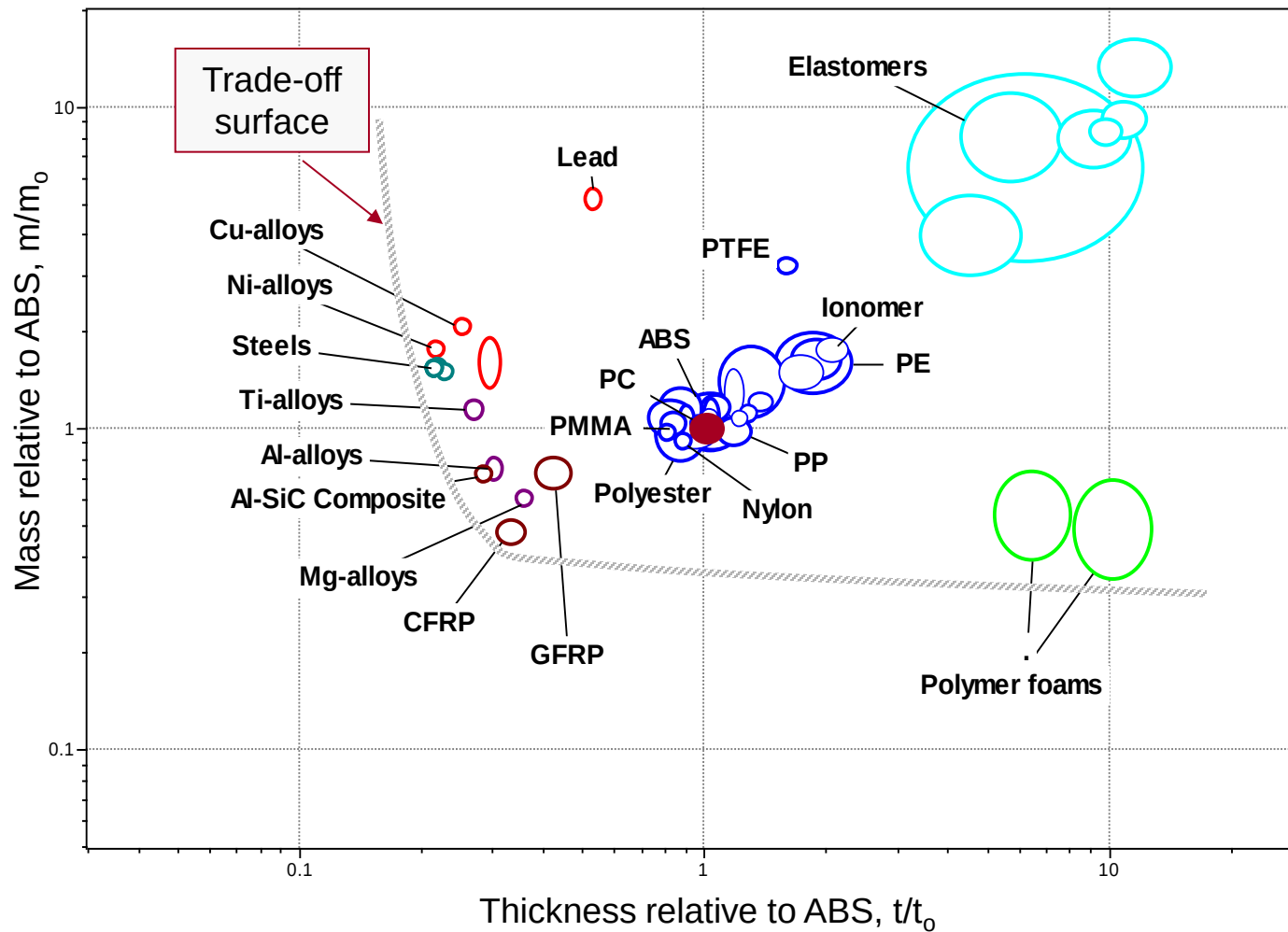
$$\frac{t}{t_o} = \left(\frac{E_o}{E} \right)^{1/3}$$

The mass differs by the factor

$$\frac{m}{m_o} = \left(\frac{\rho}{E^{1/3}} \right) \cdot \left(\frac{E_o^{1/3}}{\rho_o} \right)$$

Explore the trade-off between $\frac{t}{t_o}$ and $\frac{m}{m_o}$

The Trade-off Plot



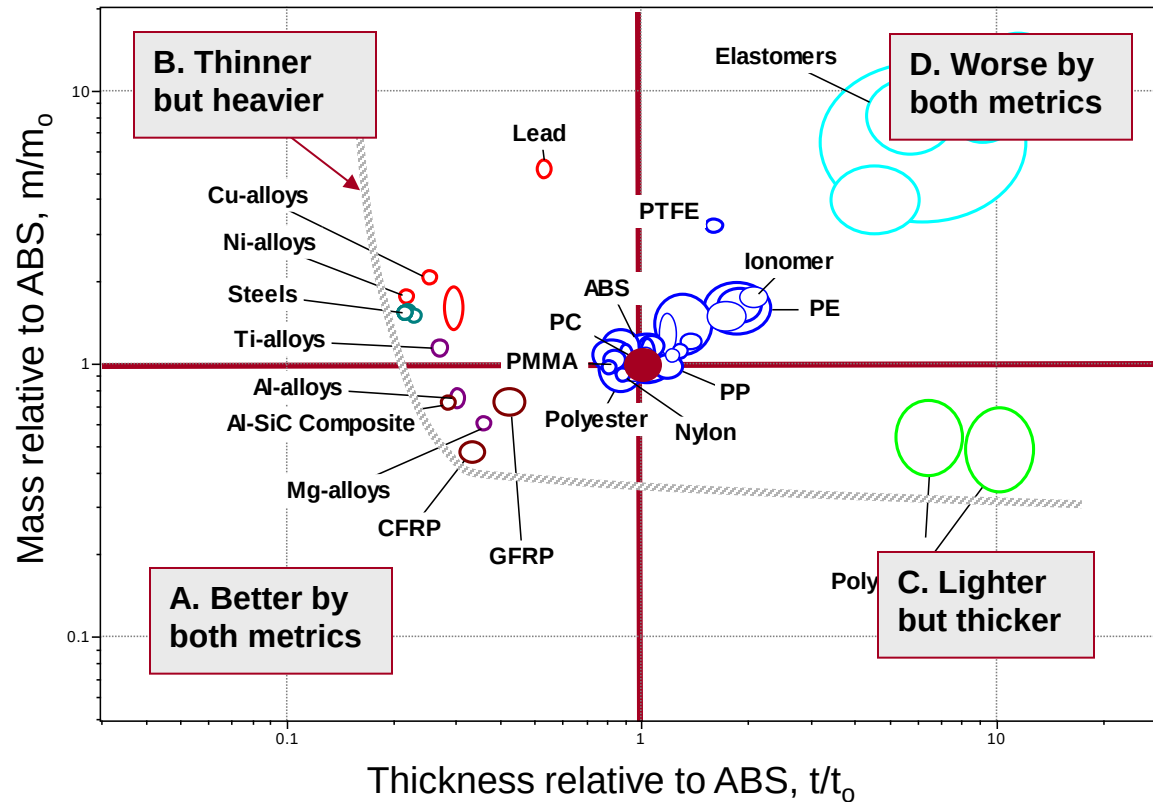
Additional constraints:

- $G_{1c} > 1 \text{ kJ/m}^2$
- Wood, leather suppressed.

- Finding a compromise: CFRP, Al and Mg alloys all offer reduction in mass and thickness

How do we work out the real cost?

- The four sectors of a trade-off plot for substitution



Is the material cost relevant? Probably not as the case only weighs a few grams. Volume and weight are much more important.